

PIN Optimization, Control, and Results

Constrained local optimization (per radio i)

$$\begin{aligned} & \max_{\pi_{\theta}} \mathbb{E} \left[\sum_{t=0}^{T-1} \gamma^t r_t \right] \\ \text{s.t. } & PDR_t^{cmd} \geq p_{\min}, L_{95,t}^{cmd} \leq \ell_{\max}, Drop_t^{cmd} \approx 0 \\ & r_t = \alpha PDR_t^{high} - \beta L_{95,t}^{high} - \delta Drop_t^{high} + \zeta Deliveries_t^{high} \end{aligned}$$

PIN local control I/O (every 250 ms)

Observation $o_t^{(i)}$: queue length by class; tx attempts/success; retries/ACK timeouts; drops/deliveries; p95 latency by class; collisions; CCA busy fraction; mean backoff.

Action $a_t^{(i)}$: for class $c \in \{\text{command, voice, best-effort}\}$

$$a_t^{(i)} = \{w_c, \eta_c, \kappa_c\}_{c=1}^3$$

where w_c is service bias, η_c admission threshold, κ_c CW scaling.

Seeded A/B demo (CSMA, 8 nodes, 1.5 s)

KPI	Base	Ctrl	Δ
High PDR	0.6995	0.6979	-0.0016
High p95 (ms)	8.9836	9.4497	+0.4662
Overall PDR	0.8193	0.8270	+0.0077
Overall p95 (ms)	36.8898	31.2521	-5.6377

Highlight seed 202: +0.0038 high-PDR,
+0.0169 overall PDR,
−5.5065 ms overall p95 latency.

